We train the policy network by updating the parameters to minimize the objective (i.e., loss) function. This involves three steps:

1. Calculate the probability of the action actually taken at each time step.
2. Multiply the probability by the discounted return (the sum of rewards).
3. Use this probability-weighted return to backpropagate and minimize the loss.

We'll look at these in turn.

Calculating the probability of the action

Calculating the probability of the action taken is easy enough. We can use the stored past transitions to recompute the probability distributions using the policy network, but this time we extract just the predicted probability for the action that was actually taken. We'll denote this quantity $P(a_t|\theta,s_t)$; this is a single probability value, like 0.75.

To be concrete, let's say the current state is S_5 (the state at time step 5). We input that into the policy network and it returns $P_\theta(A|s_5) = [0.25,0.75]$. We sample from this distribution and take action $a = 1$ (the second element in the action array), and after this the pole falls over and the episode has ended. The total duration of the episode was $T = 5$. For each of these 5 time steps, we took an action according to $P_\theta(A|s_t)$ and we stored the specific probabilities of the actions that were actually taken, $P_\theta(a|s_t)$, in an array, which might look like `[0.5, 0.3, 0.25, 0.5, 0.75]`. We simply multiply these probabilities by the discounted rewards (explained in the next section), take the sum, multiply it by –1, and call that our overall loss for this episode. Unlike Gridworld, in CartPole the last action is the one that loses the episode; we discount it the most since we want to penalize the worst move the most. In Gridworld we would do the opposite and discount the first action in the episode most, since it would be the least responsible for winning or losing.

Minimizing this objective function will tend to increase those probabilities $P_\theta(a|s_t)$ weighted by the discounted rewards. So every episode we're